

Early Prediction of Lithium-Ion Battery Degradation with a Generative Pre-trained Transformer

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ABSTRACT

Supplementary Note 1: Description of Datasets

1) MIT Dataset: The aging data from MIT¹ is derived from lithium iron phosphate (LFP)/graphite batteries. The experimental conditions involved a forced convection temperature chamber set at 30°C. The batteries have a nominal capacity of 1.1Ah and a nominal voltage of 3.3V.

All batteries in this dataset undergo a two-step fast charging strategy. The format of this strategy is expressed as "C1(Q1) – C2", where C1 and C2 represent the charging rate for the first and second constant current steps, and Q1 denotes the current switch charging step.

The first current step concludes at Q1 state of charge (SOC), and the second current step concludes at 80% SOC, after which the battery undergoes a 1C constant current-constant voltage (CC-CV) charging until reaching the upper cutoff voltage. Afterwards, maintain a constant current discharge until reaching the lower cutoff voltage. The upper and lower cutoff potentials are specified as 3.6V and 2.0V, respectively.

Supplementary Note 2: Description of Deep Learning Methods

1) Convolutional Neural Network: Convolutional Neural Networks (CNNs) are a type of deep learning model that excel in processing image data. Their architecture, inspired by the human visual cortex, is particularly effective in computer vision tasks such as image classification, object detection, and image segmentation. This makes CNNs a powerful tool not only for general image processing but also for specialized applications like predicting the SOH and EOL of batteries².

When applied to battery SOH and EOL prediction, CNNs can analyze various types of data, often converting non-image data into formats that resemble images. In this work, a CNN could process time-series data from a battery's usage history, converting this data into a 2D format akin to an image. This "image" could then be analyzed to identify patterns that are indicative of the SOH and expected lifespan of the battery³.

The CNN model, with its layers of convolutional filters, excels in identifying and extracting features from these data representations. In the context of early-stage SOH prediction, these features include charge-discharge cycles, temperature variations and voltage behaviors. By learning from historical data of batteries full lifecycle and health degradation are known, a CNN can learn to predict similar outcomes for new batteries under observation.

2) Long Short-Term Memory: Long Short-Term Memory networks (LSTMs) are a type of recurrent neural network (RNN) particularly well-suited for processing and making predictions based on time-series data. LSTMs are designed to overcome the limitations of traditional RNNs, such as the difficulty in learning long-range dependencies in sequences. This makes LSTMs highly effective for tasks involving sequential data, such as natural language processing, speech recognition, and the early-stage battery SOH prediction^{4,5}.

When applied to battery SOH and EOL prediction, LSTMs can process time-series data that reflects a power usage and condition over time. This data can include metrics like charge-discharge cycles and temperature variations that influence the SOH changes.

The key strength of LSTMs in this context lies in their ability to remember and utilize long-term dependencies in the data. For instance, they can recognize patterns and trends over extended periods, which are critical for predicting the SOH of batteries. This capability is essential for accurately forecasting how a battery will degrade under various usage conditions and environmental factors.

3) CNN and LSTM fusion model: Convolutional LSTM (Conv-LSTM)⁶ networks are an innovative integration of CNNs and Long LSTM networks. This hybrid model combines the spatial feature extraction capabilities of CNNs with the temporal sequence modeling strengths of LSTMs, making it particularly effective for tasks involving both spatial and temporal data.

In the domain of battery SOH and EOL prediction, Conv-LSTMs can effectively process and analyze data that is both spatial and temporal in nature. In this work, a Conv-LSTM could be used to analyze time-series data from the power usage history, along with spatial data that could be related to the internal structure or condition of the battery cells. This include data on charge-discharge cycles, temperature variations and voltage behaviors in this work.

4) Transformer: Transformers⁷ are a type of deep learning model that have revolutionized the field of natural language processing (NLP) and have significant potential in various other domains, including time-series data analysis for applications like predicting SOH and EOL of batteries. Unlike traditional models that process data sequentially, Transformers utilize a mechanism called "attention" to weigh the significance of different parts of the input data, allowing for parallel processing and a more nuanced understanding of data relationships.

In the context of battery SOH and EOL prediction, Transformers can analyze time-series data from a power usage history, including charge-discharge cycles, temperature variations, voltage behaviors, and other relevant parameters⁷. The model's attention mechanism allows it to focus on specific parts of the data that are most relevant for predicting the SOH and EOL, potentially identifying complex patterns and relationships that other models might miss. One of the key advantages of Transformers in this application is their ability to handle long-range dependencies in the data. This means they can effectively relate and interpret data points that are distant from each other in time.

Supplementary Note 3: Training Hyperparameters and Training Process of Deep Learning Models

In the regression prediction method, the input to the model are the charging features corresponding to three historical charge/discharge cycles. The shape of the input data varies depending on the method used. The output to the model is state of health (SOH) corresponding to the adjacent next charge/discharge cycle. The regression prediction methods used in this paper are Convolutional Neural Networks (CNN), Long Short-Term Memory Networks (LSTM), and CNN-LSTM Fusion Networks. The first 42 battery cells are used as the training set and the rest as the validation set. The parameters of the model are saved every fifty epochs, The maximum epoch number is set to 10000, and the model parameters with the lowest validation set loss are selected for validating the predictive performance of the model. All regression prediction methods use the same training parameters, the data in the training set is shuffled with a batch size of 6. and the initialised learning rate is 10e4, using the Adaptive Moment Estimation (Adam) optimiser. are trained based on a single NVIDIA RTX4090.

1) CNN: Specifically, the features of each charge/discharge cycle are reconstructed into a three-channel feature map of length and width both 60, where each channel corresponds to voltage, current and temperature, respectively. Subsequently, the feature maps corresponding to three historical charge/discharge cycles are stacked together to form the input data to the CNN. Such a shape of the input data helps to preserve the time-series information of battery aging and allows the CNN to efficiently capture the time dependence. CNN for regression prediction consist of several key components:

- **Convolutional layers:** two convolutional layers are used in this paper and these layers are used to extract aging features at different levels. The first convolutional layer is responsible for capturing the correlation between various battery features in each charge-discharge cycle. The number of input channels is 3, corresponding to the channels of the reconstructed feature map, the number of output channels is 1, the convolution kernel size is (3,3), the stride is 1 and the padding is 1. The output feature matrix of the first convolutional layer is of the same length and width as the input feature map, and the number of channels is 1. The feature matrices of the three charge/discharge cycles are stacked to form a new three-channel feature map as the input to the second convolutional layer to capture temporal correlations between the three charge/discharge cycles and deeper aging features. The number of input channels is 3, corresponding to the number of historical charge/discharge cycles. The rest of the convolution parameters are the same as in the first convolutional layer.
- **Flatten Layer:** the output of the convolutional layers is passed through the flatten layer to transform the multidimensional data into one-dimensional vectors for input to the linear layer.
- **Linear layers:** the CNN model used for regression prediction contains two linear layers. The first linear layer has an input feature dimension of 3600 and an output feature dimension of 100 while the second linear layer has an input feature dimension of 100 and an output feature dimension of 1. These linear layers are used to generate the final SOH predictions.

2) LSTM: Each charge/discharge cycle generates a large amount of battery features that encompasses the electrochemical and physical properties of the battery, as well as the effects of ageing. However, due to the large number of features, placing all the features of a charge/discharge cycle in a single time node to use the more interpretable LSTM structure may face computational and parametric problems. To overcome this problem, in this paper, battery features from three historical charge/discharge cycles are reconstructed into a feature sequence of length 20, where each time step of the sequence is a set of 1620-dimensional features, and such a sequence was fed into the LSTM model. The LSTM model consists of the following key components:

- **LSTM layers:** this paper uses three-layer LSTM, each with a hidden layer dimension of 25. This allows the model to capture temporal dependencies and sequence patterns at multiple levels of abstraction. The number of input features for each layer is 1620, which is the number of features in the reconstructed charging feature sequence.
- **Flatten layer:** the output of the LSTM layer is flattened into a one-dimensional vector. This step is done to convert the output of the LSTM into a shape that fits the input of the linear layer.
- **Linear layer:** the flatten feature vectors are fed into a linear layer which has an input feature dimension of 500 and an output feature dimension of 1. This linear layer is used to generate the final SOH predictions.

3) CNN-LSTM: The input to the CNN-LSTM network is the same as the CNN described in the previous section, in which the features of each charge/discharge cycle are reconstructed into a three-channel feature map according to the same pattern. This type of data processing helps to preserve the time-series information and the dynamics of the battery features. However, instead of using a flatten layer, CNN-LSTM reconstructs the aged features obtained through two convolutional layers into a feature sequence of length 20, which preserves the temporal structure of the data and passes it into the LSTM layer for further temporal modelling. The CNN-LSTM model consists of the following key components:

- **Convolutional layer:** the convolutional layer hyperparameters of CNN-LSTM networks are the same as those of CNN networks, which are two layers of convolutional layers.
- **LSTM layer:** since the input data of the LSTM layer is changed, the number of input features of the LSTM layer is 180, and the other parameters of the LSTM layer are set the same as the previous LSTM layer.
- **Flatten layer:** the output of the LSTM layer is flattened into a one-dimensional vector. This step is done to convert the output of the LSTM into a shape that fits the input of the linear layer.
- **Linear layer:** the flatten feature vectors are fed into a linear layer which has an input feature dimension of 500 and an output feature dimension of 1. This linear layer is used to generate the final SOH predictions.

Supplementary Note 4: Training parameters and training process of the self-regression model

The autoregressive prediction method is divided into two main modules according to the function. The first module uses the time-series features of early-stage prediction starting offset (EPSO) as input to predict the time-series features of full lifecycle. The second module takes the features of each cycle as input and estimates the SOH of the corresponding cycle. Therefore, the autoregressive prediction model takes the EPSO's time-series features as input to obtain the SOH prediction results of full lifecycle. In this paper, the autoregressive prediction model is discussed in the following sections.

Time-series features prediction module

The time-series features of EPSO are used as input to the time-series features prediction module. The inputs are turned into time-series tokens of EPSO after tokenization. The vocabulary size of each token is the same as BatteryGPT, which is 743. The output of the time-series features prediction module is the next token of the time-series, which is reconverted to the next feature of the time-series after the opposite tokenization rule. The first 42 battery cells are used as the training set and the rest as the validation set. The training is terminated when the autoregressive error, root mean square error, of the validation and training sets is less than 5% or the epoch number reaches 500000. The minimum epoch number is set to 1000. All time-series features prediction modules in this paper are using the Adaptive Moment Estimation (Adam) optimizer and are trained based on a single NVIDIA RTX4090.

LSTM

In this paper, time-series features represent continually gathered battery features over a specific period, typically reflecting the evolving trends of battery properties. LSTM can analyze historical battery performance feature to predict the battery's feature at future time points. The initialized learning rate of the LSTM is 10-4, as the number of epochs increases, the learning rate will decrease to 10-5 at the 4000th epoch. All input time-series features are divided by a window of length 256 with a batch size of 64. The LSTM for time-series features prediction consists of the following key components:

- Embedding layer: the main role of this layer is to embed tokens into multidimensional features, conforming to the input of the LSTM layer. In this paper, vocabulary size is 743, and dimension of output embedding is set to 128.
- LSTM layer: the number of input features is 128, the hidden layer dimension is 2000, and the number of layers in the LSTM layer is 6.
- Flatten layer: the output of the LSTM layer is flattened into a one-dimensional vector. This step is done to convert the output of the LSTM into a shape that fits the input of the linear layer.
- Linear layer: the dimension of input features is 512000 and the dimension of output features is 743, which is same as vocabulary size.
- Classification layer: sampling from a probability distribution via softmax and Multinomial Distribution (MD) to determine the next feature to be generated.

Transformer

The initialized learning rate of the Transformer is 10-6, as the number of epochs increases, the learning rate will decrease to 10-7 at the 4000th epoch. All input time-series features are divided by a window of length 256 with a batch size of 2. The Transformer for time-series features prediction consists of the following key components:

- Embedding layer: the main role of this layer is to embed tokens into multidimensional features, conforming to the input of the Transformer layer. In this paper, vocabulary size is 743, and dimension of output embedding is set to 128.
- Transformer layer: The number of input features is 1024, the number of layers for encoder and decoder is the same, both are 16, and the dimension of the hidden layer of the feedforward neural network (FFN) is $1024^2 = 1048576$. The role of FFN is to introduce non-linear mappings, enabling the model to learn more complex functional relationships. This helps enhance the model's representation capabilities of input data, allowing it to better capture patterns and features within input sequences.

- Linear layer: the dimension of input features is 512000 and the dimension of output features is 743, which is same as vocabulary size.
- Classification layer: sampling from a probability distribution via softmax and Multinomial Distribution (MD) to determine the next feature to be generated.

SOH Estimation Module

In the SOH estimation module, the input data to the module are the charging features corresponding to a single charge/discharge cycle, and the shape of the input will vary depending on the method used. The output of the model is the SOH corresponding to the corresponding charge/discharge cycle. The first 42 battery cells are used as the training set and the rest as the validation set. The model's parameters are saved every fifty epochs, and the training process is set to run for a maximum of 10,000 epochs. The model parameters associated with the lowest validation set loss are chosen to assess the predictive performance of the model during validation. The estimation methods all use the same training parameters, the data in the training set is shuffled with a batch size of 6. The initial learning rate for all estimation methods is set to 0.0001, and the Adam optimizer is utilized across all methods. Training is conducted on a single NVIDIA RTX4090.

CNN

The features of each charge-discharge cycle are reconstructed into three-channel feature maps with a length and width both set to 60. Each channel corresponds to voltage, current, and temperature, respectively. CNN for regression prediction consist of several key components:

- Convolutional layers: two convolutional layers are used in this paper and these layers are used to extract aging features at different levels. The number of input channels of the first convolutional layer is 3, the number of output channels is 6, the convolutional kernel size is (3,3), the step size is 1 and the padding is 1. The number of input channels of the second convolutional layer is 6, the number of output channels is 1, and the rest of the parameters are the same.
- Flatten layer: the output of the convolutional layer is flattened into a one-dimensional vector to fit the input of the linear layer.
- Linear layers: the CNN model used for SOH estimation contains two linear layers. The first linear layer has an input feature dimension of 3600 and an output feature dimension of 100 while the second linear layer has an input feature dimension of 100 and an output feature dimension of 1. These linear layers are used to generate the final SOH prediction.

LSTM

In this paper, battery features are reconstructed into a feature sequence of length 20, where each time step of the sequence is a set of 540-dimensional features, and such a sequence was fed into the LSTM model. The LSTM model consists of the following key components:

- LSTM layers: this paper uses three-layer LSTM, each with a hidden layer dimension of 25. The number of input features for each layer is 420, which is the number of features in the reconstructed charging feature sequence.
- Flatten layer: the output of the LSTM layer is flattened into a one-dimensional vector. This step is done to convert the output of the LSTM into a shape that fits the input of the linear layer.
- Linear layer: the flatten feature vectors are fed into a linear layer which has an input feature dimension of 500 and an output feature dimension of 1.

CNN-LSTM

In this paper a fusion network structure is introduced that combines CNN with LSTM network to deal with battery features. The input to the CNN-LSTM network is the same as the CNN described in the previous section, in which the features of each charge/discharge cycle are reconstructed into a three-channel feature map according to the same pattern. However, instead of using a flatten layer, CNN-LSTM reconstructs the aged features obtained through two convolutional layers into a feature sequence of length 20, which preserves the temporal structure of the data and passes it into the LSTM layer for further temporal modelling. The CNN-LSTM model consists of the following key components:

- Convolutional layer: the convolutional layer hyperparameters of CNN-LSTM networks are the same as those of CNN networks, which are two layers of convolutional layers.
- LSTM layer: since the input data of the LSTM layer is changed, the number of input features of the LSTM layer is 180, and the other parameters of the LSTM layer are set the same as the previous LSTM layer.
- Flatten layer: the output of the LSTM layer is flattened into a one-dimensional vector. This step is done to convert the output of the LSTM into a shape that fits the input of the linear layer.
- Linear layer: the flatten feature vectors are fed into a linear layer which has an input feature dimension of 500 and an output feature dimension of 1.

Multilayer Perceptron

The Multilayer Perceptron (MLP) is a type of feedforward neural network composed of one or multiple hidden layers, with each hidden layer containing multiple neural nodes. MLP is widely used in machine learning tasks such as classification and regression. In this paper, the SOH is estimated by inputting the battery features after flattening them into a two-layer MLP with an input layer dimension of 10800, a hidden layer dimension of 150, and an output layer dimension of 1.

Supplementary Note 5: Hyperparameter Optimization of BatteryGPT

Hyperparameter Optimization (HPO) is a crucial step in the training of machine learning models. In the context of machine learning, hyperparameters are external configuration settings for a model that are not learned from the data but must be specified before the training process begins. Examples include learning rates, regularization strengths, and the number of hidden layers in a neural network. The goal of hyperparameter optimization is to find the optimal set of hyperparameters that maximizes the performance or accuracy of a model on a specific task. There are various techniques for hyperparameter optimization, including grid search, random search, and more sophisticated algorithms such as Bayesian optimization and genetic algorithms. These methods systematically explore the hyperparameter space to identify the combination that leads to the best model performance.

Due to the limitation of the experimental equipment, this paper uses the method of grid search to find the optimal hyperparameters. The search hyperparameter space is as follows:

batch_size: Represents the number of samples in each training batch, with only one choice, which is **4**.

attn_drop_ratio: Represents the dropout ratio in the attention mechanism, with three choices: 0.5, **0.25**, and 0.1.

block_size: Represents the number of previous characters in each input sequence, with three choices: **256**, 512, and 1024.

depth: Represents the depth of the decoder, with only one choice, which is **5**.

embedding: Represents the size of the embedding layer, with three choices: **384**, 768, and 1024.

Learning_rate: Represents the learning rate taken during the learning process, with two choices: **1e-3** and 1e-4.

Supplementary Note 6: Evaluation on the KIT Dataset

The KIT dataset⁸ is introduced to assess the external validity of BatteryGPT beyond the MIT benchmark used in the main text. KIT contains long-horizon cycling under EV-like operating profiles, multiple ambient temperatures, and pronounced accelerated-degradation knees. This note summarizes the evaluation protocol, baselines, metrics, and results on KIT, and explains why they substantiate the claims made for BatteryGPT.

Dataset and preprocessing Cells are cycled over long durations with periodic check-ups and temperature variation. We adopt the same tokenization scheme used for MIT: voltage, current, and temperature are discretized into non-overlapping vocabularies with fixed offsets; per-dataset numeric ranges are estimated from the training split with a small safety buffer. All experiments use chronological splits. The SOH estimator operates on short windows of generated tokens, identical to the design described in the main text.

Evaluation protocol BatteryGPT is prompted with the initial portion of a life and then run in a free-running, fully autoregressive mode to the end of life. We report two prompt lengths: BatteryGPT(5) and BatteryGPT(30), where the number denotes the number of observed cycles provided as the prompt before free generation. This protocol matches the early-prediction use case. For interpretability and the missing-data analysis, we also consider predictive starting offsets (EPSO) as illustrated in Supplementary Fig. S5: with a fixed prompt length (256 tokens), the model can start at 5%, 10%, or 30% of life and generate the remainder; this is equivalent to late starts or long missing prefixes.

	Method	SOH Prediction RMSE (%)	SOH Prediction MAPE (%)	Knee Point Error	Knee Point MAPE (%)	EOL Error	EOL MAPE (%)
Regressive prediction	CNN	7.41	7.27	98	9.62	Out of range	Out of range
	LSTM	12.41	15.58	-240	23.57	Out of range	Out of range
	CNN-LSTM	2.83	2.85	33	3.24	38	3.25
Autoregressive prediction	LSTM	11.94	13.48	-265	25.95	Out of range	Out of range
	Transformer	22.71	30.56	-421	41.07	Out of range	Out of range
	BatteryGPT (5)	3.63	2.59	-62	6.83	-12	1.23
	BatteryGPT (30)	2.28	2.01	-34	3.74	-3	0.37

Supplementary Table 1. Comparison of LIB degradation prediction performance for KIT dataset. The out of range means that the predicted SOH do not reach 80%, and EOL can not be obtained.

Baselines We compare against (i) regressive predictors trained to map early cycles directly to SOH: CNN, LSTM, and CNN-LSTM; and (ii) native autoregressive forecasters (LSTM and Transformer) run free-form over long horizons. All baselines use the same splits and early-prompt regime; hyperparameters are tuned on validation data.

Results On KIT, BatteryGPT achieves accurate long-horizon trajectory prediction and downstream prognostics in Supplementary Table. 1:

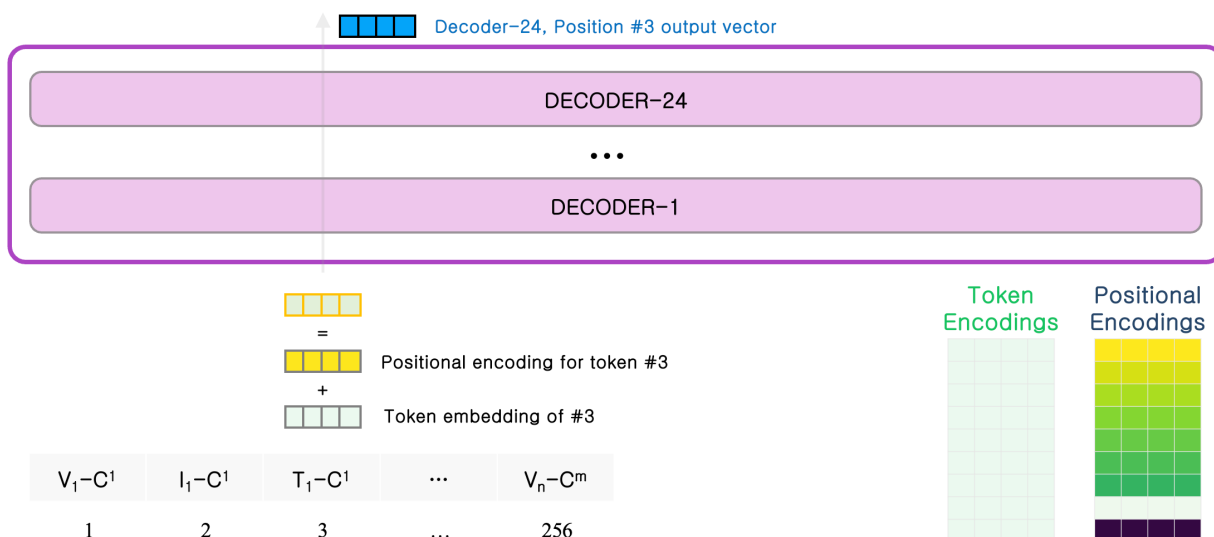
- BatteryGPT(5) (EPSO = 5 cycles): RMSE 3.63%, MAPE 2.59%, KP Error -62 cycles, KP MAPE 6.83%, EOL Error -12 cycles, EOL MAPE 1.23%.
- BatteryGPT(30) (EPSO = 30 cycles): RMSE 2.28%, MAPE 2.01%, KP Error -34 cycles, KP MAPE 3.74%, EOL Error -3 cycles, EOL MAPE 0.37%.

These results compare favorably with the strongest regressive baseline CNN-LSTM (RMSE 2.83%, MAPE 2.85%, KP Error 33 cycles, KP MAPE 3.24%, EOL MAPE 3.25%) and are markedly better than naive long-horizon autoregressive LSTM/Transformer forecasters (RMSE 11.94–22.71%, MAPE 13.48–30.56%), which accumulate exposure bias when run free-form.

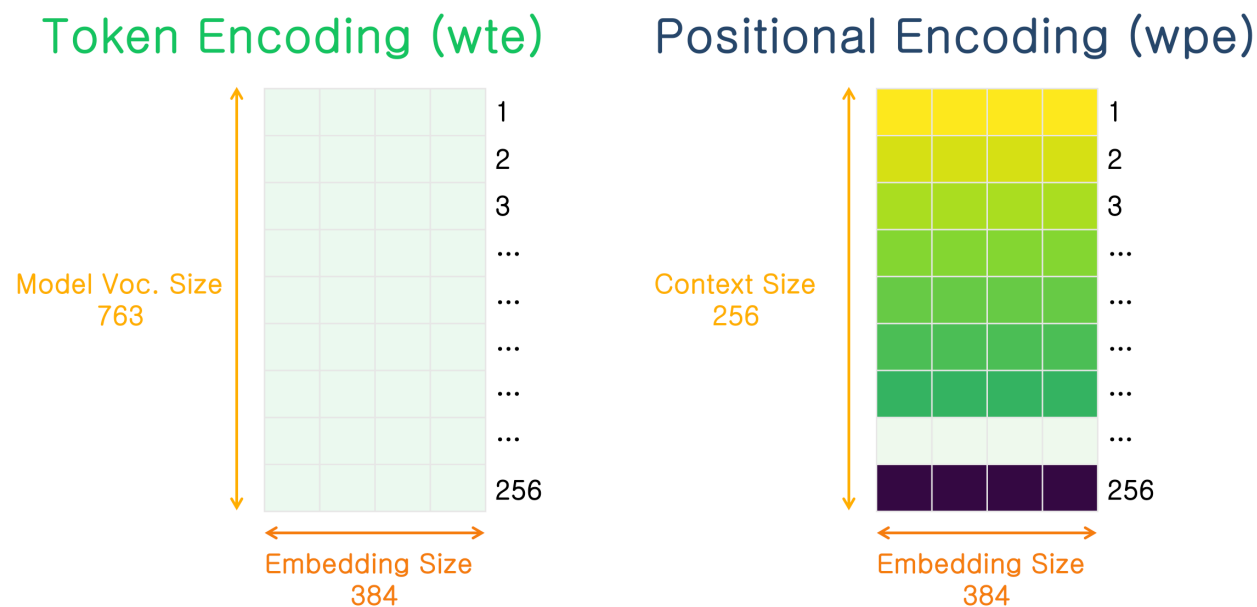
Supplementary Figures:



BatteryGPT



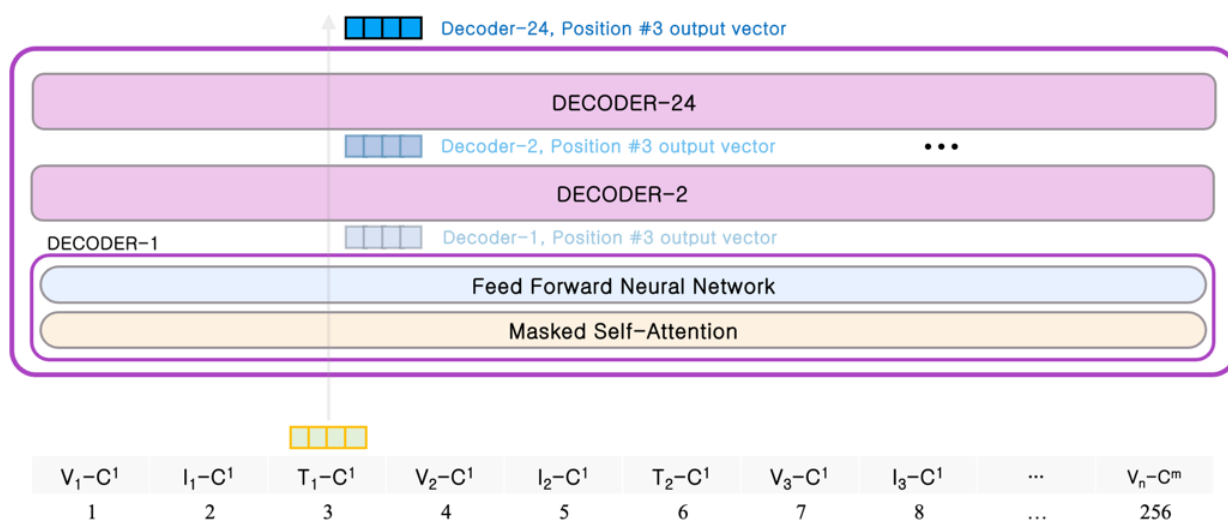
Supplementary Figure S1. The diagram shows the hierarchical structure of the 24-layer decoder stack, where each decoder layer processes input embeddings combined with positional encodings (Supplementary Fig. S2). The process starts with the token embedding of a voltage token (such as #1), which is then enhanced by adding the positional encoding for a specific token position (in this case, token #3). This combined input is processed sequentially through the decoders from 1 to 24, resulting in the output vector from the 24th decoder layer for the specified token position.



Supplementary Figure S2. This diagram illustrates the structures of token encoding (wte) and positional encoding (wpe). The token encoding matrix (left) has dimensions of 763 (model vocabulary size) by 384 (embedding size), where each cell represents a specific token's embedding. The positional encoding matrix (right) has dimensions of 256 (context size) by 384 (embedding size), with each cell encoding positional information used to maintain the order of tokens within the context.



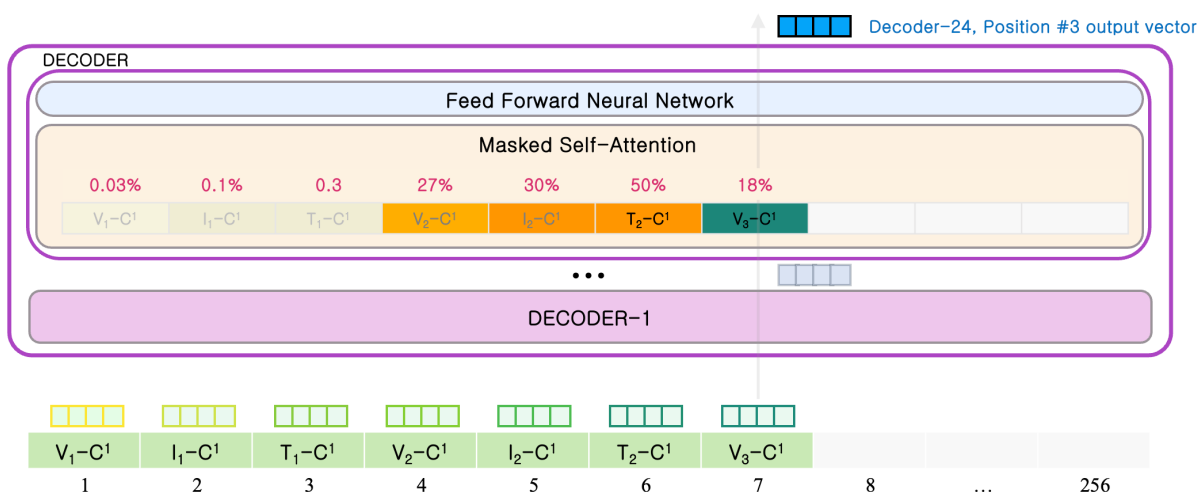
BatteryGPT



Supplementary Figure S3. This diagram provides a detailed view of the model's decoder architecture and its data flow. The diagram highlights the 24-layer decoder stack, with a focus on the output vectors at each layer for a specific token position (#3). The decoding process begins with masked self-attention, followed by a feed-forward neural network within each decoder layer. As the token embeddings progress through each layer, starting from decoder 1 up to decoder 24, the model refines its understanding of the token. The figure emphasizes the sequential processing of input embeddings, showing how the initial token and positional embeddings are transformed through each decoder layer to produce the final output vector.



BatteryGPT



Supplementary Figure S4. This diagram illustrates the detailed mechanism of masked self-attention within the decoder layer. It shows how different tokens in the input sequence, each represented by their token embeddings, are weighted during the attention process. The percentages indicate the attention weights assigned to various tokens (e.g., V1-C1, I1-C1, etc.) when processing the token at position #3 (T3-C1). These weights reflect the importance of each token in predicting the next token in the sequence. The output of the masked self-attention mechanism is then passed through a feed-forward neural network within the decoder layer. The diagram highlights the flow from the initial input embeddings to the final output vector from decoder-24 for token position #3.



BatteryGPT



Supplementary Figure S5. Predictive starting points and data utilization of BatteryGPT under different EPSo settings. Taking a battery with a lifecycle of 600 cycles as an example. When EPSo = 0%, the input to the model consists of 256 tokens starting from the 1st cycle. When EPSo = 10%, the input to the model consists of 256 tokens starting from the 60th cycle. When EPSo = 30%, the input to the model consists of 256 tokens starting from the 180th cycle.

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